

Philip (Phi) Bui

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EDUCATION

- Ph.D. in Urban Systems, New York University (NYU), New York, NY** Sep 2026 - Present
- Advised by Dr. Debra Laefer. Research: honest uncertainty in infrastructure sensing, from measurement to agent.
- M.S. in Computer Science: Human-Robot Interaction, Tufts University, Medford, MA** Sep 2024 - May 2025
- Graduated with a 3.8/4.0 GPA.
- B.S. in Computer Science, Worcester Polytechnic Institute (WPI), Worcester, MA** Aug 2021 - Dec 2023
- Graduated with a 3.8/4.0 GPA and distinction.

SKILLS

- Languages: Python, TypeScript/JavaScript, SQL, C++, C#, Java, Bash.
- ML: PyTorch, TensorFlow, agentic workflows, (multi-agent) reinforcement learning, LLM evaluation.
- Infrastructure: AWS, Kubernetes, Docker, Cloudflare, PostgreSQL, Qdrant, Node.js, React.

PUBLICATIONS & RESEARCH

- State-Wise Constrained Policy Shaping for Zero-Shot Runtime Behavior Steering** 2026
- Second author, with T. Howell, R. McPherson, and Dr. V. Sarathy. Peer-reviewed workshop paper, AAAI-26 AIGOV workshop, Singapore. Developed a post-training supervisor that enforces safety norms on DQN agents in HighwayEnv. Collisions fell 97% in-distribution and 99% zero-shot.
- Scoring Rules Certify Reporting, Not Seeking: A Dilemma for Level-Based Rewards in Sequential, Embodied Perception** 2026
- Sole author. Level-based proper scoring-rule rewards certify honest reporting but not information-seeking, for every strictly proper binary rule. Poster, IROS-26 Full-Shift Robot Co-Workers workshop (non-archival).
- Do Agents Disclose What the Data Cannot Support? A Live-Data Benchmark for Abstention, Measurement-Uncertainty Caveats, and Missingness Disclosure over Municipal Open Data** 2026
- Sole author. A 596-question live-data benchmark over a municipal open-data API.
 - Claude Haiku 4.5 quotes the documented reliability caveat on 100% of answers that need one, yet discloses a discoverable data defect in 0 of 108 eligible answers. Released: github.com/phbui/civic-honesty-benchmark.
- Gestalt: Hybrid Retrieval and Long-Term Memory for AI Agents** (personal project, 2025 - Present)
- Built hybrid BM25 and dense-embedding retrieval with a temporal knowledge graph over 500+ entries. A golden-set benchmark tracks recall@5 and MRR, and the tests fail on regression.

RESEARCH EXPERIENCE

- Swarm Robotics AI Research Intern, Air Force Research Laboratory, Eglin Air Force Base, FL** May - Aug 2025
- With Dr. Scott Nivison. Designed heterogeneous multi-agent RL algorithms for swarm coordination under communication constraints. (Classified; details omitted.)
- Graduate Research Assistant, Human Interaction Lab, Tufts University, Medford, MA** Jan - May 2025
- With Dr. Saul Albert, Dr. Umair Muhammad, and Dr. J.P. de Ruiter. Led the back-end overhaul of GailBot, a PyPI-distributed transcription pipeline for conversation-analytic research. Engineered the REST APIs and the PostgreSQL schema.

PROFESSIONAL EXPERIENCE

- Full-Stack Software Engineer, Cyvl, Somerville, MA** Aug 2025 - Aug 2026
- Extended production pipelines over petabyte-scale georeferenced imagery and LiDAR point clouds that detect pavement distress, signs, markings, and above-ground assets across road networks.
 - Built a visual embedding pipeline indexing 500k+ street-level images per city into a Qdrant vector store for sub-second semantic retrieval.
 - Designed the AI ingestion pipeline that converts municipal engineering documents into geospatial files. It became Cyvl's winning submission to NYC DEP's Environmental Technology Lab Challenge.
- Software Engineer, enLabel, Boston, MA** Jan 2024 - Jan 2025
- Fixed hundreds of defects in legacy C#/ASP.NET code and led a React/TypeScript UI redesign. Authored an end-to-end regression-testing suite for the Web API.